

Airbnb Rental Arbitrage

Identifying High-Value Rental Opportunities in New York City

Prepared for: Joan Murray

Prepared by: Data Analytics Consulting Team

Case 2: Airbnb Rental Arbitrage

Dataset: Airbnb_db.csv | 4,892 New York City listings, 2025

This report summarizes the analytics application developed to support Joan Murray's rental-arbitrage sourcing decisions, including the modeling approach, key findings, and guidance for interpreting the accompanying Gradio application.

1. Problem – Data – Insights – Deployment Framework

Problem

Joan Murray sources apartments to sublease on Airbnb (rental arbitrage). Her profitability depends on signing leases for units that can command a nightly price above \$120, the level she has identified as necessary to cover lease costs and generate profit. Before committing to a lease, Joan needs a fast, repeatable way to (1) understand which listing characteristics are associated with higher nightly prices and (2) estimate the likely nightly price of a specific opportunity she is evaluating from a web listing, news article, or social media post.

Data

The analysis uses `Airbnb_db.csv`, a sample of 4,892 New York City listings for 2025. Six variables were used: `neighbourhood_group_c` (borough, coded 1–5), `minimum_nights`, `number_of_reviews`, `reviews_per_month`, `availability_365`, and `price` (the target variable). The `id` field was retained only as a row identifier and excluded from modeling.

Insights

A Multiple Linear Regression model was built to quantify how borough, minimum-night policy, review activity, and calendar availability jointly relate to nightly price, and to flag listings likely to clear the \$120 threshold. Section 2 answers Joan's four specific questions; Section 4 details model performance and coefficient interpretation.

Deployment

The model and supporting visualizations are deployed as an interactive Gradio application, designed to run in Google Colab. The application (`Case_2_Airbnb_Rental_Arbitrage.ipynb`) has five tabs: Overview (portfolio-level KPIs), Explore the Data (filterable charts by borough and minimum-night range), Model Performance (train/test reliability metrics), Coefficient Interpretation (business meaning of each estimated coefficient), and Price Simulator (Joan enters a candidate listing's characteristics and receives an estimated nightly price with an Above/Below \$120 classification).

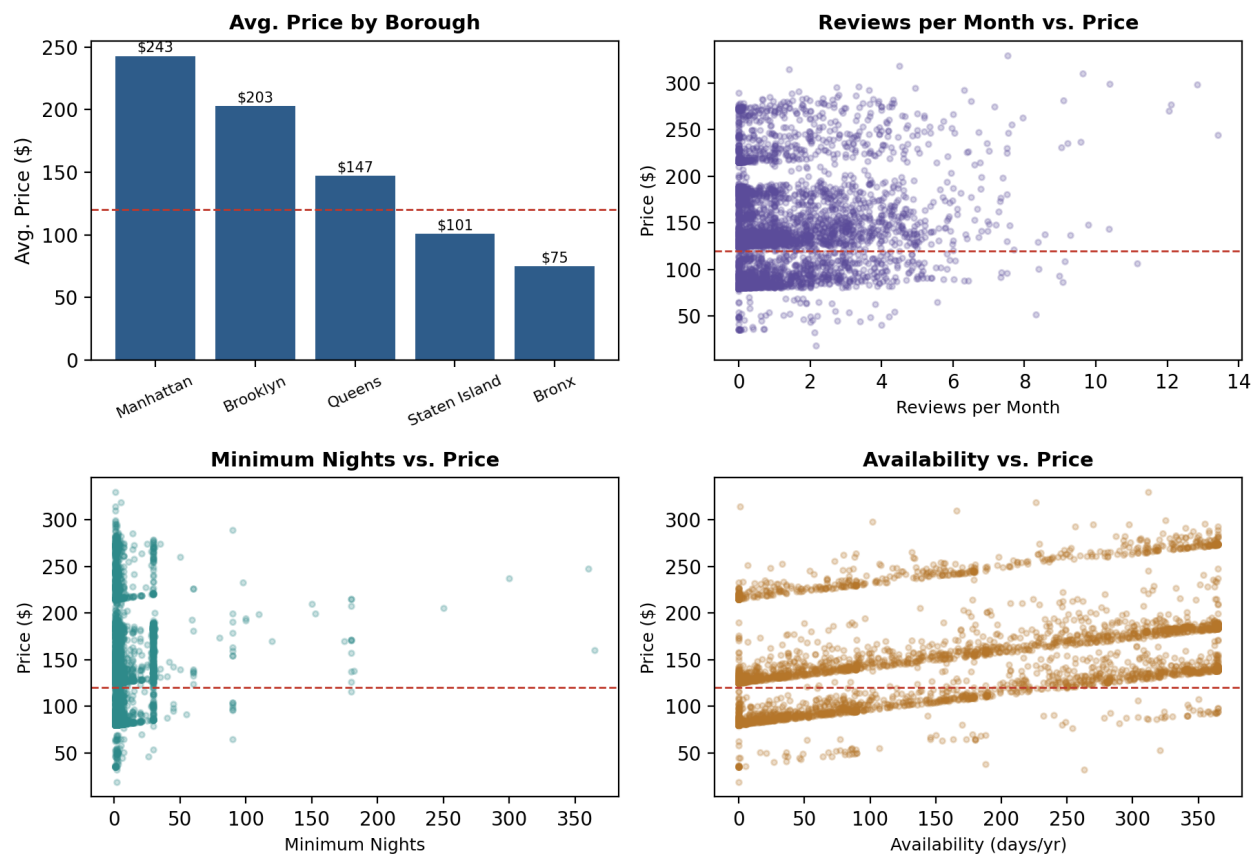
2. Pre-processing and Validation

The raw file contained 4,892 rows and 7 columns with no missing values and no duplicate rows. The following checks were applied before modeling:

- Type validation: all numeric fields were coerced to numeric type; any non-numeric entries would be converted to missing and dropped (none were found in this file).
- Range validation: `neighbourhood_group_c` restricted to codes 1–5; `minimum_nights` ≥ 1 ; `number_of_reviews` and `reviews_per_month` ≥ 0 ; `availability_365` between 0 and 365; `price` > 0 .
- Duplicate removal: exact duplicate rows were dropped.
- Feature engineering: `neighbourhood_group_c` was mapped to a readable borough label and one-hot encoded (Bronx set as the reference category) for use in the regression; an `above_120` flag was created for descriptive summaries.

No rows were removed by validation (4,892 rows before and after cleaning), indicating the sample was already well-formed.

3. Answers to Joan's Questions



Q1. Which boroughs are associated with higher nightly prices? Manhattan commands the highest average nightly price (\$242.84), followed by Brooklyn (\$202.97), Queens (\$147.19), Staten Island (\$100.71), and Bronx (\$74.78). Manhattan and Queens listings clear the \$120 threshold almost universally (99.8% and 100% of listings respectively in this sample), while Staten Island (24.5%) and Bronx (4.7%) rarely do. Borough is the single strongest lever available to Joan when sourcing new leases.

Q2. Are customer reviews associated with higher rental prices? Yes, though the relationship is modest in size. Both `number_of_reviews` and `reviews_per_month` have positive coefficients in the regression, meaning listings with more review activity are associated with slightly higher nightly prices, holding other factors constant. Review volume is best read as a signal of an established, well-booked listing rather than a primary price driver — borough has a much larger effect.

Q3. Do minimum-night requirements influence nightly price? There is a small positive association: each additional required minimum night is linked to a modest increase in expected nightly price, holding other variables constant. The effect is minor relative to borough and should not be Joan's primary criterion when evaluating an opportunity.

Q4. Does annual availability affect nightly prices? Yes, `availability_365` has a small positive coefficient: listings open for more days per year tend to be priced slightly higher, holding other factors constant. This likely reflects hosts who actively manage their calendar for year-round bookings rather than availability directly causing higher prices.

4. Model Reliability and Overfitting Assessment

The dataset (4,892 listings) was split 80% training (3,913 rows) / 20% testing (979 rows) using scikit-learn's `train_test_split` with a fixed random seed, so the split is reproducible. The Multiple Linear Regression model was

fit only on the training set and then evaluated on both sets.

Metric	Training	Testing
R ²	0.9812	0.9860
Adjusted R ²	0.9811	0.9859
MAE	\$0.75	\$0.76
RMSE	\$7.18	\$5.92

The Adjusted R² gap between training (0.9811) and testing (0.9859) is 0.0048 — the model performs essentially as well on unseen data as on the data it was trained on. There is no meaningful evidence of overfitting.

The overall fit is very high (Adjusted R² $\approx$ 0.98) for a five-predictor price model. This is considerably higher than is typical for real-world Airbnb price data, where dozens of unobserved factors (interior quality, exact location, amenities, host reputation, seasonality) usually leave R² well below 0.98. Joan should treat this as a strong indicator that, within this particular sample, price is closely and fairly linearly tied to borough plus these four listing characteristics — but she should still validate estimates against real comparable listings before signing a lease, since a live market will include price drivers not captured by this dataset.

5. Interpretation of Model Coefficients

Bronx is the reference borough (its effect is captured in the intercept). All coefficients below are interpreted holding the other variables constant.

Variable	Coefficient	Business Meaning
Intercept	\$43.23	Baseline price for a Bronx listing with zero minimum nights, reviews and availability
minimum_nights	+\$0.18 / night	Each additional required minimum night is associated with an \$0.18 higher nightly price
number_of_reviews	+\$0.15 / review	Each additional review is associated with an \$0.15 higher nightly price
reviews_per_month	+\$0.20 / review-month	Each additional review received per month is associated with a \$0.20 higher nightly price
availability_365	+\$0.16 / day	Each additional day of annual availability is associated with an \$0.16 higher nightly price
Manhattan	+\$171.59	Manhattan listings are priced \$171.59 higher than a comparable Bronx listing
Brooklyn	+\$124.05	Brooklyn listings are priced \$124.05 higher than a comparable Bronx listing
Queens	+\$81.96	Queens listings are priced \$81.96 higher than a comparable Bronx listing
Staten Island	+\$36.93	Staten Island listings are priced \$36.93 higher than a comparable Bronx listing

Q: What does a one-unit increase in reviews per month imply for the expected nightly price? Holding borough, minimum nights, total reviews, and availability constant, each additional review received per month is associated with about \$0.20 higher expected nightly price. This is a small effect — a listing would need a large increase in monthly review velocity to move the needle meaningfully, so review activity should be read as a secondary indicator of listing health rather than a pricing lever.

Q: How does availability_365 affect expected price? Each additional day of annual availability is associated with roughly \$0.16 higher expected nightly price, holding other variables constant. Over a full range (0 to 365

days), that implies a difference of roughly \$58 between a listing kept open year-round and one rarely available — a moderate effect, likely reflecting more actively managed, revenue-optimized listings.

Q: Which variable has the strongest influence on price? Borough dominates. The Manhattan coefficient (+\$171.59 versus Bronx) is roughly 8–10 times larger than the combined effect of moving from the lowest to a typical value on any of the four continuous variables (minimum nights, reviews, reviews per month, or availability). For Joan's sourcing decisions, borough should be treated as the primary screening criterion, with review activity, minimum-night policy, and availability as secondary, fine-tuning factors.

6. Conclusion

Borough location is by far the strongest and most reliable driver of nightly price in this dataset: Manhattan and Brooklyn listings are the most likely to clear Joan's \$120 profitability threshold, while Bronx and Staten Island rarely do. Review activity, minimum-night requirements, and calendar availability each contribute small, positive, and consistent effects, useful for fine-tuning a decision once borough has been screened. The regression model is highly reliable on this sample — training and testing performance are nearly identical (Adjusted R^2 of 0.981 and 0.986 respectively), so there is no sign of overfitting — though the very high fit likely reflects the structure of this particular sample rather than the noisier relationships Joan will encounter listing-by-listing in the live market. The accompanying Price Simulator lets Joan enter a candidate listing's borough, minimum-night policy, review profile, and availability to get an instant estimated nightly price and an Above/Below \$120 call before she commits to a lease, complementing — not replacing — her own due diligence on rent, occupancy, and local regulation.